6.8 Piecewise Functions

Algebra II 6.8 · Student sheet

Name _____ Period ____ Date _____

I can…

- evaluate piecewise functions.
- graph and write piecewise functions.
- write step functions and absolute value functions as piecewise functions.

Part 1 · Phone plan

A phone plan costs $30 a month for up to 5 GB of data, plus $10 for each GB over 5.

GB used	1	2	3	4	5	6	7	8
Cost ($)								

Graph the points (let each box up be $10).

Rule for $0 \le g \le 5$: _____ Rule for $g > 5$: _____

Part 2 · Notes: fill in the blanks

A **piecewise function** uses different _____ on different parts of the _____.

To evaluate, first find which piece's _____ the input fits.

≤ or ≥ → _____ circle. < or > → _____ circle.

A **step function** has pieces that are all _____.

Part 3 · Examples (work them with the class)

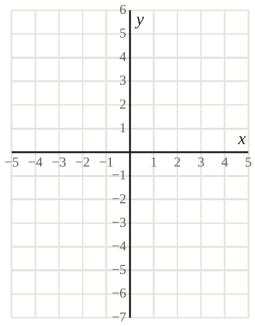
Example 1

Let $f(x) = 2x + 1$ if $x \le 1$, and $f(x) = x^2 - 3$ if $x > 1$.

Find $f(-2)$, $f(1)$, and $f(3)$.

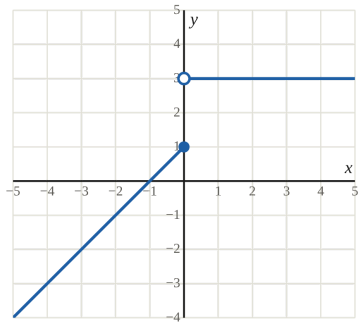
Example 2

Graph f(x) = x + 3 if x < 1, and f(x) = −2x + 1 if x ≥ 1.



Example 3

Write a piecewise function for the graph.



Example 4

A parking garage charges $5 for the first hour or part of an hour and $3 for each additional hour or part of an hour, up to 3 hours.

Write a step function for the cost. How much is 2.5 hours?

Example 5

Write $y = |x - 3|$ as a piecewise function.

Example 6

Write the phone plan from the opening as a piecewise function $C(g)$. Find $C(8)$.

Part 4 · Partner practice

Partner 1

Let $f(x) = -x$ if $x < -1$; 4 if $-1 \le x \le 2$; x^2 if $x > 2$.

Find $f(-3)$, $f(2)$, and $f(5)$.

Partner 2

Write $y = |x + 1|$ as a piecewise function.